Interactive Simulation of the Chernobyl Disaster

A Simplified Model of an RBMK-1000 Reactor

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Abstract

This report describes an interactive Python simulation of an RBMK nuclear reactor developed as an educational model of the physical processes associated with the Chernobyl accident. The simulation is not intended to reproduce the reactor with engineering precision. Instead, it implements a simplified Monte Carlo particle model in which neutrons move through a two-dimensional reactor core and interact with uranium fuel, graphite moderator, water coolant, steam voids, xenon poison and boron control rods. The main objective is to show how microscopic neutron interactions can produce macroscopic reactor behaviour such as changing neutron population, reactivity, coolant boiling and power instability.

1 Introduction

The Chernobyl accident is one of the most significant examples of unstable feedback in a nuclear reactor. The RBMK reactor design combined graphite moderation with water cooling. This design allowed the reactor to remain moderated even when water in the core began to boil. As a result, steam formation could reduce neutron absorption without removing the graphite moderator, producing a positive void coefficient.

The aim of this project was to create a simplified but interactive model of this behaviour. The simulation allows the user to observe the development of a chain reaction, manipulate control rods, and monitor changing diagnostic quantities. The focus is educational: the model should make the main reactor-physics mechanisms visible and understandable, rather than provide a validated accident reconstruction.

The simulator was inspired by a visual particle-based representation of the Chernobyl sequence. The final version extends this idea into a Python application with a reactor-core view, independently adjustable control rods, a diagnostics panel and time-dependent plots.

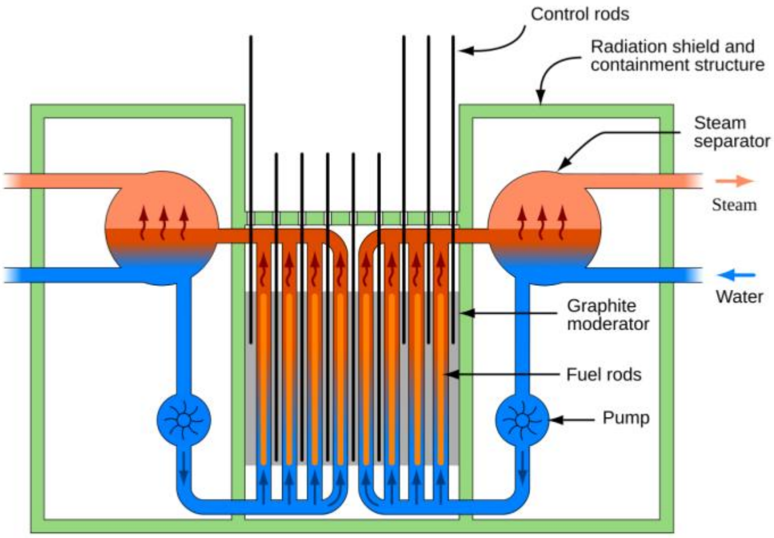

Figure 1: Simplified RBMK reactor schematic showing fuel rods, graphite moderator, control rods, water circulation, pumps and steam separators.

2 RBMK Reactor Background

The simplified reactor model is based on several essential features of an RBMK-type reactor. The core contains uranium fuel, where fission occurs after neutron absorption. Graphite acts as the main moderator and slows fast neutrons to thermal energies. Water removes heat from the reactor, but it also absorbs neutrons. Boron control rods absorb neutrons and reduce reactor power. Xenon-135 acts as a strong neutron poison and can suppress the chain reaction after previous operation.

In this simulation, the reactor state is determined mainly by neutron balance. If each generation of neutrons produces, on average, exactly enough new neutrons to sustain the reaction, the reactor is critical. If too many neutrons are absorbed or lost, the system becomes subcritical. If neutron production exceeds losses, the reactor becomes supercritical.

The simulator therefore focuses on the following mechanisms:

- fission chain reaction in uranium fuel,
- graphite and water moderation,
- neutron absorption by coolant, xenon and control rods,
- heat production in fuel channels,
- boiling of coolant into steam,
- positive void feedback,
- estimation of k_{eff} and reactivity.

3 Monte Carlo Particle Model

3.1 Neutron Motion

Neutrons are represented as individual particles moving through a two-dimensional reactor core. Each neutron has a position and velocity. Between interactions, neutron motion is updated using a first-order Euler step:

$$x(t+\Delta t) = x(t) + v_x \Delta t, \qquad y(t+\Delta t) = y(t) + v_y \Delta t. \tag{1}$$

To reduce the probability of missing collisions, the timestep is subdivided into smaller integration steps:

$$n_{\mathrm{sub}} = \left\lceil \frac{v_{\mathrm{fast}} \Delta t}{3} \right\rceil, \qquad \Delta t_{\mathrm{sub}} = \frac{\Delta t}{n_{\mathrm{sub}}}. \tag{2}$$

This method is not a full neutron transport solver. It is a simplified real-time approximation designed to give visually interpretable particle motion while preserving the main qualitative effects of neutron multiplication and absorption.

3.2 Stochastic Interaction Rule

The simulation treats neutron interactions as stochastic events. For a process with rate λ, the probability of an event in a short interval Δt is approximated by

$$P(\text{event}) = \lambda \Delta t, \qquad \lambda \Delta t \ll 1. \tag{3}$$

A random number $u \sim U(0,1)$ is drawn for each possible interaction. The event occurs if

$$u < \lambda \Delta t. \tag{4}$$

This rule is used for moderation, absorption, xenon decay, fuel regeneration and similar processes. The approach makes the simulation suitable for a real-time graphical application because the microscopic events are simple to compute but still produce non-deterministic reactor behaviour.

3.3 Fission and Neutron Multiplication

When a neutron collides with a reactive uranium fuel element, fission may occur. Thermal neutrons have a much higher probability of causing fission than fast neutrons. In the simplified simulation, a fission event produces three main effects:

- new fission-born neutrons,
- heat deposited into the local fuel channel,
- production of xenon-related poison material.

The emitted neutrons can then trigger further fissions, forming a chain reaction. If absorption and leakage are small enough, the neutron population grows. If absorption by control rods, coolant or xenon dominates, the neutron population decreases.

4 Xenon Poisoning

Xenon-135 is an important part of the simulated accident physics. It has a very large neutron absorption effect and can strongly suppress the chain reaction. In the model, xenon is represented by poison particles that absorb thermal neutrons. Xenon can also decay or be removed by neutron absorption.

This produces delayed feedback. When xenon concentration is high, many neutrons are absorbed before they can cause fission. The reactor therefore has reduced power. If the neutron flux later rises, xenon is burned away. This removes a strong absorber and can allow the neutron population to increase quickly.

This behaviour is important for understanding why a reactor can appear unresponsive at low power and then become unstable after control rods are withdrawn and xenon is depleted.

5 Coolant, Temperature and Positive Void Feedback

5.1 Void-Dependent Absorption

The main RBMK instability represented in the simulation is the positive void coefficient. Water coolant absorbs neutrons, but steam absorbs fewer neutrons. When the coolant boils, neutron absorption decreases. Because graphite moderation remains present, more neutrons can survive and cause fission.

The effective coolant absorption rate is modelled as

$$\lambda_c = \lambda_{fc}(1 - f_v) + \lambda_{rod} n_{rod}, \tag{5}$$

where λ_{fc} is the full-water coolant absorption rate, f_v is the steam void fraction, λ_{rod} is the absorption contribution of a neighbouring inserted control rod and n_{rod} is the number of relevant neighbouring rods.

The qualitative feedback is therefore:

$$f_v \uparrow \quad \Rightarrow \quad \lambda_c \downarrow \quad \Rightarrow \quad \text{neutron absorption} \downarrow \quad \Rightarrow \quad \text{reactivity} \uparrow . \tag{6}$$

This is the central mechanism that allows boiling to increase power in the simplified model.

5.2 Channel Temperature

The reactor is divided into fuel channels. Each channel has a local fission flux ϕ_i, which increases when fissions occur and decays over time:

$$\phi_i(t + \Delta t) = \phi_i(t) \exp\left(-\frac{\Delta t}{\tau_\phi}\right). \tag{7}$$

The coolant temperature in channel i is updated by a simple heat-balance equation:

$$T_i(t + \Delta t) = T_i(t) + H\phi_i(t)\Delta t - CT_i(t)\Delta t, \qquad T_i \geq 0. \tag{8}$$

Here, H represents heat deposited by fission and C represents cooling.

5.3 Steam Void Fraction

Steam void fraction is calculated from the channel temperature using a clamped linear relation:

$$f_{v,i} = \text{clamp}\left(\frac{T_i - T_{\text{boil}}}{T_{\text{full}} - T_{\text{boil}}}, 0, 1\right). \tag{9}$$

The average steam void fraction displayed in the diagnostics panel is

$$\bar{f}_v = \frac{1}{N_{\text{ch}}} \sum_{i=1}^{N_{\text{ch}}} f_{v,i}. \tag{10}$$

6 Software Implementation

The simulation was implemented as an interactive Python program. The graphical interface contains three main parts: a reactor-core view, a plots and diagnostics panel, and a control panel.

The user can change the insertion of control rods and observe the response of the reactor in real time.

Figure 2: Final simulation interface showing the RBMK core, control rods, fuel channels, diagnostics and user controls.

The implementation can be divided into several subsystems:

- **Particle engine**: updates neutron position, speed and collisions.
- **Reaction engine**: handles fission, moderation, absorption and xenon interactions.
- **Thermal model**: computes fission flux, channel temperature and steam void fraction.
- **Control-rod model**: applies boron absorption according to rod insertion depth.
- **Diagnostics**: estimates neutron count, k_{eff}, reactivity and void fraction.
- **User interface**: draws the core, plots and control panel.

The visual model makes the reactor state directly observable. Blue particles represent uranium/fuel-related material, small moving particles represent neutrons, grey or dark regions represent moderator and structure, and vertical rods represent control rod insertion. The diagnostic panel complements the visual core by showing numerical estimates of the reactor state.

7 Reactor Diagnostics

Because individual particle interactions are noisy, raw event counts are smoothed before being shown to the user. For a diagnostic quantity x, the exponentially smoothed value x_e is computed as

$$x_e(t + \Delta t) = x_e(t)\alpha + x(t), \qquad \alpha = \exp\left(-\frac{\Delta t}{\tau_{\text{EMA}}}\right). \tag{11}$$

The effective multiplication factor is estimated from the smoothed neutron balance:

$$k_{\text{eff}} = \frac{N^e_{\text{fiss-n}}}{N^e_{\text{fiss}} + N^e_{\text{loss}}}. \tag{12}$$

Here, $N^e_{\text{fiss-n}}$ is the smoothed number of fission-born neutrons, N^e_{fiss} is the smoothed number of fission events and N^e_{loss} is the smoothed number of neutrons lost through absorption or leakage.

Reactivity is computed as

$$\rho = \frac{k_{\text{eff}} - 1}{k_{\text{eff}}}, \tag{13}$$

and expressed in dollars using

$$\rho[\$] = \frac{\rho}{\beta_{\text{eff}}}, \qquad \beta_{\text{eff}} = 0.0065. \tag{14}$$

The reactor is classified as subcritical, critical or supercritical according to the sign and magnitude of the reactivity.

8 Simulation Results

The simulator produces time-dependent plots that show how the reactor responds to user actions and internal feedback. The neutron population plot shows rapid increases when fission dominates and decreases when neutron losses dominate.

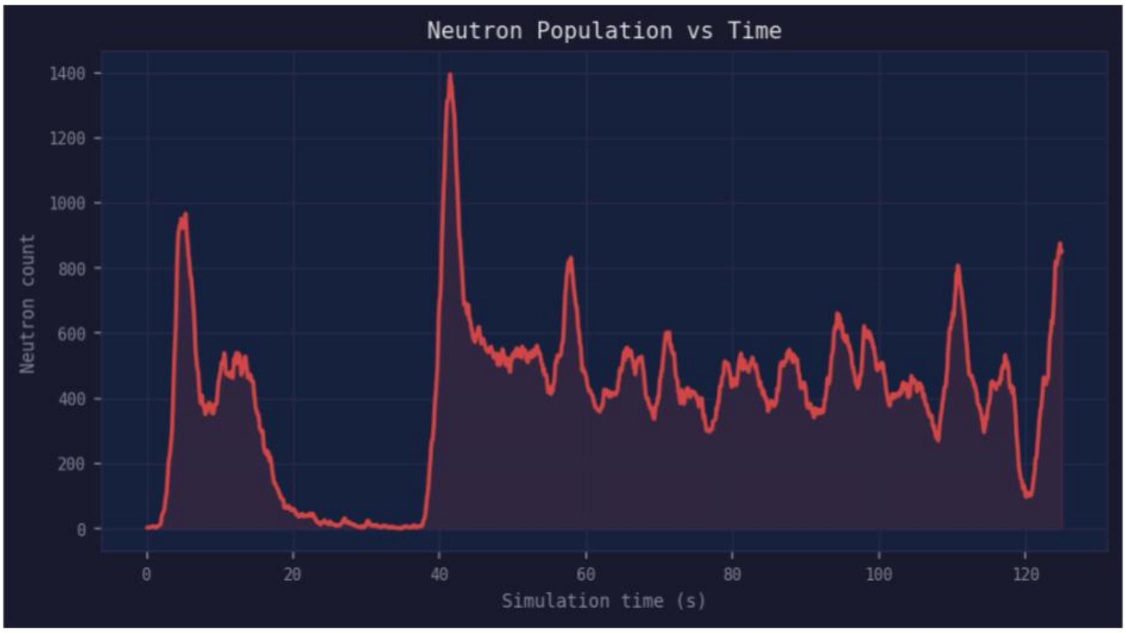

Figure 3: Neutron population as a function of simulation time. Peaks correspond to periods of increased fission activity.

The k_{eff} plot shows whether the reactor is subcritical, critical or supercritical. Values near $k_{\text{eff}} = 1$ correspond to approximately critical behaviour. Values above one indicate that neutron multiplication exceeds losses.

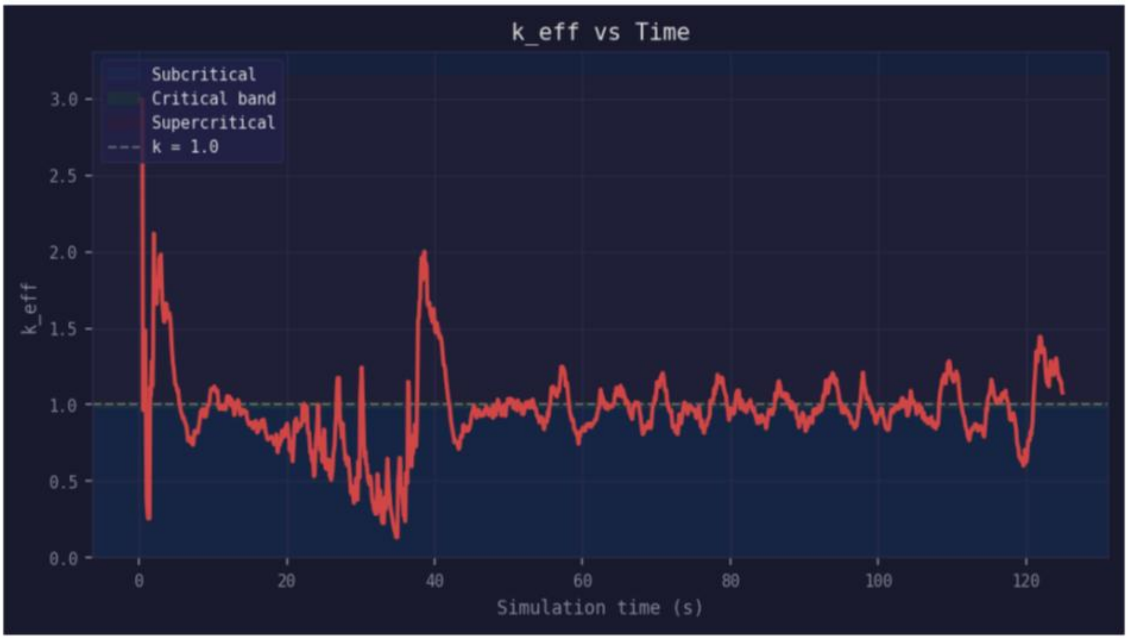

Figure 4: Estimated effective multiplication factor k_{eff} over time. The dashed reference line marks $k_{\text{eff}} = 1$.

The steam void plot shows the development of boiling in the reactor channels. When the void fraction exceeds the danger threshold, coolant absorption is significantly reduced and the reactor becomes more sensitive to power excursions.

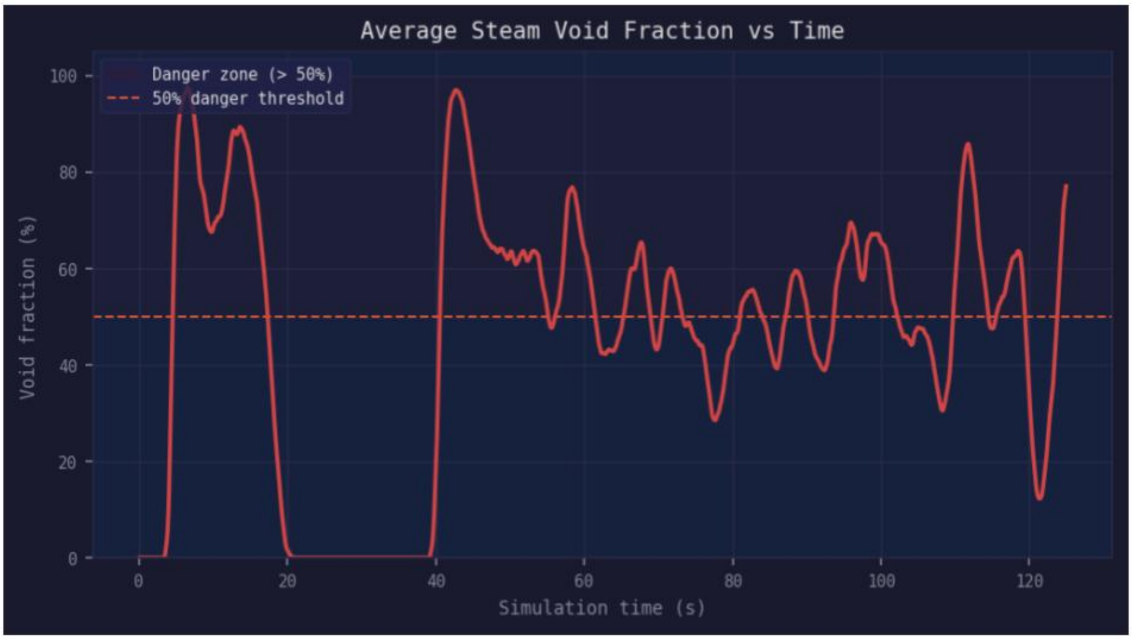

Figure 5: Average steam void fraction over time. The dashed line marks the selected danger threshold.

The final channel-temperature plot shows non-uniform heating across the reactor. Some channels exceed the full-void threshold, meaning that they contribute strongly to the positive void feedback.

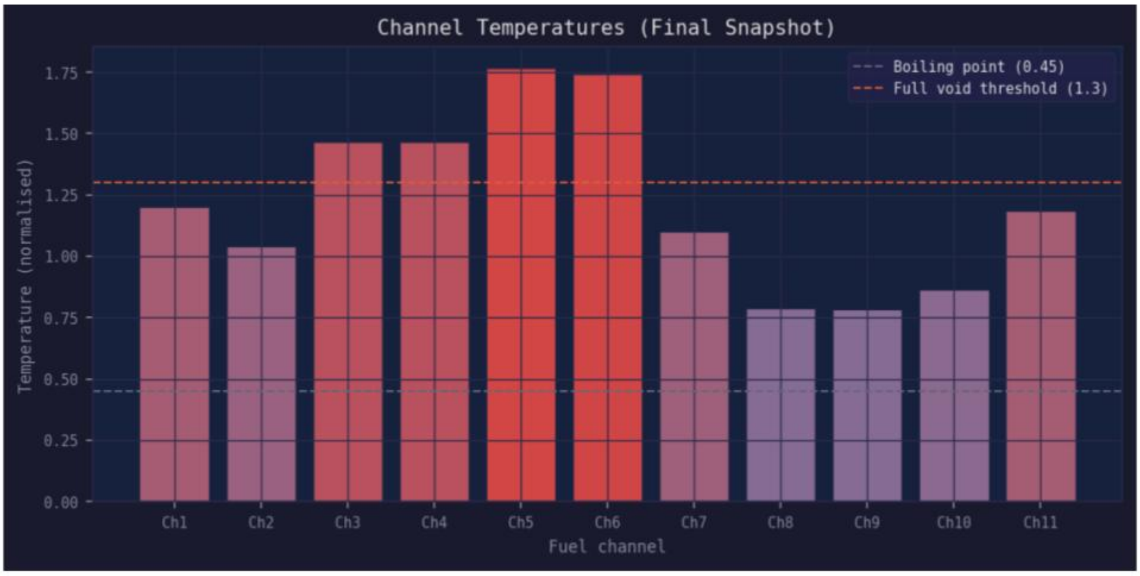

Figure 6: Final channel temperatures. The dashed lines indicate the boiling point and full-void threshold used by the simplified model.

The plots show that the model can reproduce several qualitative reactor behaviours:

- control rod withdrawal increases neutron survival,
- increasing fission raises coolant temperature,
- boiling reduces coolant absorption,
- reduced absorption increases reactivity,
- xenon poisoning can suppress power and later disappear through burn-off,
- the reactor can move between subcritical, critical and supercritical states.

9 Model Parameters

The simulation uses both physically motivated values and tuned parameters. The effective delayed-neutron fraction $\beta_{\text{eff}} = 0.0065$ is taken from published reactor-physics data. The number of neutrons emitted per fission is treated as approximately two to three. Other parameters are adjusted to produce clear qualitative behaviour in real time.

Table 1: Selected constants used in the simplified simulation model.

Symbol	Description	Value
β_{eff}	Effective delayed-neutron fraction	0.0065
ν	Neutrons emitted per fission	2–3
P_{th}	Thermal fission probability	0.92
P_{fast}	Fast fission probability	0.05
λ_G	Graphite moderation rate	9.0 s^{-1}
$\lambda_{W,\text{mod}}$	Water moderation rate	1.2 s^{-1}
$\lambda_{W,\text{abs}}$	Water absorption rate	1.7 s^{-1}
λ_{fc}	Full-water coolant absorption rate	1.9 s^{-1}
λ_{rod}	Per-rod boron absorption worth	8.5 s^{-1}
λ_{Xe}	Xenon decay rate	0.04 s^{-1}
T_{boil}	Boiling threshold	0.45
T_{full}	Full-void threshold	1.3

These values are not intended to be a validated numerical description of an actual RBMK-1000 reactor. They are used to make the reduced model stable, visible and interactive.

10 Limitations

The model has several major limitations. The reactor geometry is two-dimensional, while a real RBMK core is a complex three-dimensional system. Neutron motion is represented using simple particle kinematics rather than full neutron transport. Thermal hydraulics are reduced to local channel temperatures and linear steam void formation. The model does not include detailed delayed-neutron precursor groups, fuel burnup, pressure dynamics, material deformation, structural failure or the complete AZ-5 control-rod insertion sequence.

The simulation also stops before physical destruction of the reactor. It models the development of unstable neutron and thermal feedback, not the mechanical explosion itself. Therefore, the simulator should be interpreted as an educational physics sandbox, not a reactor safety tool or accident-analysis code.

11 Conclusion

This project demonstrates that a simplified Monte Carlo particle model can be used to explain important reactor-physics mechanisms associated with the Chernobyl accident. The simulator connects microscopic events, such as neutron absorption and fission, with macroscopic quantities such as neutron population, steam void fraction, channel temperature, reactivity and k_{eff}.

The final application provides an interactive environment where the user can manipulate control rods and observe how neutron multiplication, xenon poisoning and positive void feedback affect reactor stability. Although the model is simplified and not quantitatively predictive, it is useful as a visual and computational teaching tool for understanding how reactor instability can emerge from coupled physical feedback mechanisms.

References

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